

REINVENT-Based Generative Design of Factor Xa Inhibitors Using Scaffold-Aware ML Scoring

Project Summary

This project demonstrates an end-to-end generative molecular design workflow for **Factor Xa (FXa) inhibitor discovery** using **REINVENT4 transfer learning** and a previously developed **scaffold-aware machine learning activity model**.

The goal was to test whether a REINVENT prior model could be shifted toward generating molecules with higher predicted FXa activity after transfer learning on potent FXa-like molecules. The result was positive: even a short 5-epoch transfer-learning sanity run enriched the generated chemical space for molecules with higher predicted FXa pKi.

Main Result, Corrected for Unequal Sample Sizes

The original prior and transfer-learned (TL) model produced different numbers of valid unique molecules after sampling: **982** for the prior and **911** for the TL model. Therefore, the most defensible comparison uses both raw counts and rates normalized by the number of valid molecules.

Metric	Original REINVENT Prior	FXA Transfer-Learned Model	Interpretation
Valid generated molecules	982	911	TL produced fewer valid unique molecules after sampling.
Mean predicted pKi	6.531	6.674	+0.143 pKi shift.
Median predicted pKi	6.509	6.647	+0.138 pKi shift.
Max predicted pKi	7.922	8.942	+1.020 pKi increase.
Top-10 mean predicted pKi	7.573	8.021	+0.448 pKi enrichment among top-ranked molecules.
Predicted pKi ≥ 7	102 / 982 (10.4%)	171 / 911 (18.8%)	+69 molecules; +8.4 percentage points.
Predicted pKi ≥ 8	0 / 982 (0.0%)	4 / 911 (0.4%)	TL produced high-scoring molecules absent from the prior sample.
Novel vs FXA reference	982 / 982 (100.0%)	911 / 911 (100.0%)	Both sets were fully novel by flat canonical SMILES comparison.
Basic druglike molecules	791 / 982 (80.5%)	600 / 911 (65.9%)	Druglike fraction decreased by 14.7 percentage points.

Key interpretation: Transfer learning increased the fraction of generated molecules with predicted pKi ≥ 7 from **10.4%** to **18.8%**, while generating **4** molecules with predicted pKi ≥ 8 compared with **0** from the original prior. This is a rate-normalized enrichment, not just a raw-count artifact.

The main trade-off was drug-likeness: the basic druglike fraction decreased from **80.5%** to **65.9%**, which justifies the downstream candidate-prioritization step.

Candidate Prioritization Result

After scoring the TL-generated molecules, candidates were filtered for:

1. Valid RDKit molecule
2. Novel relative to the Project 1 FXA reference set
3. Basic druglike properties
4. High predicted FXA pKi

This produced **600 valid, novel, druglike TL-generated candidates**. Among these 600 candidates, **92 had predicted pKi ≥ 7** and were assigned to three priority tiers:

Candidate Tier	Criteria	Count
Tier 1	Novel + druglike + predicted pKi ≥ 8.0	2
Tier 2	Novel + druglike + predicted pKi 7.5-8.0	13
Tier 3	Novel + druglike + predicted pKi 7.0-7.5	77
Docking queue	Top prioritized candidates for structure-based follow-up	30

The best prioritized candidate from the docking queue was:

Property	Value
SMILES	<chem>O=c1ccccc1-c1ccc(NS(=O)(=O)c2ccc(Cl)s2)cc1</chem>
Predicted FXA pKi	8.116
RF tree uncertainty	1.290
MW	366.85
cLogP	3.35
QED	0.769
Novel vs FXA reference	Yes
Basic druglike flag	Yes

Project Context

This project builds directly on Project 1, where a scaffold-aware Factor Xa activity model was trained using curated ChEMBL FXa bioactivity data.

Project 1 model used here:

```
models/fxa_05b_best_scaffold_feature_model.joblib
```

The scoring model was a scikit-learn pipeline:

```
SimpleImputer -> VarianceThreshold -> RandomForestRegressor
```

The model was trained on Morgan fingerprints:

```
Morgan radius = 2
Fingerprint size = 2048 bits
Binary Morgan fingerprint
```

In this project, the model was used as a scoring function to evaluate molecules generated by REINVENT.

Workflow Overview

Project 1 FXA ML model

|

```

      v
Prepare potent FXA seed molecules
      |
      v
Flatten and clean SMILES for REINVENT transfer learning
      |
      v
Filter unsupported elements for REINVENT prior compatibility
      |
      v
Split into train/validation SMILES
      |
      v
Run REINVENT transfer learning
      |
      v
Sample molecules from original prior and TL model
      |
      v
Score generated molecules with FXA RF model
      |
      v
Compare prior vs TL score distributions with rate-normalized metrics
      |
      v
Prioritize novel, druglike, high-scoring candidates
      |
      v
Create docking queue for structure-based validation

```

Data Preparation

Potent FXa molecules were selected from the curated Project 1 dataset using $pK_i \geq 8.0$. The initial potent seed set contained **1249** molecules. After removing stereochemical notation and deduplicating flat canonical SMILES, **1211** unique flat potent FXa SMILES remained.

The first TL launch identified molecules containing unsupported elements for the REINVENT prior. A robust RDKit atom-symbol filter was applied using the supported elements C, N, O, S, F, Cl, and Br.

Metric	Count
Input flat SMILES	1211
Supported SMILES retained	1199
Removed SMILES	12
Drop fraction	0.99%

Unsupported elements removed were B (9 molecules), I (2 molecules), and P (1 molecule).

Transfer Learning Setup

REINVENT4 was used in transfer-learning mode.

```

run_type = "transfer_learning"
device = "cpu"

```

```

num_epochs = 5
save_every_n_epochs = 1
batch_size = 50
sample_batch_size = 1000

input_model_file = "external/REINVENT4/priors/reinvent.prior"
output_model_file = "models/fxa_reinvent_tl_pki8_all_flat_sanity.model"

smiles_file = "data/reference/tl_split/fxa_tl_train_pki8_all_flat_elements.smi"
validation_smiles_file = "data/reference/tl_split/fxa_tl_valid_pki8_all_flat_elements.smi"

standardize_smiles = false
randomize_smiles = true
randomize_all_smiles = false

```

The final split contained approximately **959 training SMILES** and **240 validation SMILES**. The short transfer-learning run completed successfully, with the best validation loss observed at epoch 5. Because this was a short CPU sanity run, the result should be interpreted as a proof of concept rather than a fully optimized production model.

Sampling and Scoring

Two 1000-molecule sampling runs were performed for fair comparison:

1. Original REINVENT prior
2. FXA transfer-learned REINVENT model

Generated molecules were scored using the Project 1 Factor Xa RandomForest model. Morgan fingerprints were generated with RDKit `MorganGenerator` using the same settings as Project 1: radius 2, 2048 bits, binary fingerprint, no chirality, and bond types enabled. RF uncertainty was estimated as standard deviation across tree predictions after applying the fitted preprocessing steps.

Key Output Files

```

results/tables/project2_final_key_metrics.csv
results/metrics/project2_final_summary.json
results/metrics/project2_final_summary.txt
results/figures/prior_vs_tl_key_metrics.png
results/figures/prior_vs_tl_predicted_pki_distribution.png
results/figures/tl_candidate_tier_counts.png
results/tables/tl_fxa_tier1_pki8_druglike_candidates.csv
results/tables/tl_fxa_tier2_pki7p5_druglike_candidates.csv
results/tables/tl_fxa_top100_ranked_candidates.csv
results/tables/tl_fxa_docking_queue_top30.csv

```

The most important practical output is:

```
results/tables/tl_fxa_docking_queue_top30.csv
```

This file contains the top candidates for structure-based validation.

Reproducible Script Order

```

01_prepare_reinvent_seed_smiles.py
02_score_generated_smiles_with_fxa_model.py
03_prepare_tl_flat_seed_smiles.py
04a_make_tl_train_valid_split.py

```

04b_filter_tl_smiles_by_reinvent_supported_elements.py
05_score_tl_generated_smiles_with_fxa_model.py
06_score_prior_generated_smiles_with_fxa_model.py
06_compare_prior_vs_tl_scores.py
07_prioritize_tl_fxa_candidates.py
08_make_project2_summary_figures.py

Strengths

This project is strong as a portfolio piece because it connects curated bioactivity modeling, scaffold-aware ML scoring, generative molecular design, transfer learning, novelty checking, drug-likeness filtering, uncertainty-aware prioritization, and a docking-ready output queue.

It shows a complete computational drug discovery workflow from activity model to generative design to candidate triage.

Limitations

This project does not claim experimental activity.

Important limitations:

1. Predicted pKi values are model-based prioritization scores, not measured affinities.
 2. The TL run was intentionally short and CPU-based, so it should be treated as a proof-of-concept sanity run.
 3. The RandomForest model has uncertainty, especially for novel generated molecules.
 4. Druglike fraction decreased after TL, showing why prioritization filters are necessary.
 5. Docking and structure-based validation are still required.
 6. ADMET and synthetic accessibility filters were not yet fully integrated.
 7. No experimental validation was performed.
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Next Steps

The next logical project stage is structure-based validation.

Recommended follow-up:

1. Convert the top-30 docking queue to 3D conformers.
 2. Prepare a Factor Xa receptor structure.
 3. Dock candidates into the FXa active site.
 4. Compare docking poses with known FXa inhibitor interactions.
 5. Filter by docking score, binding mode, key interactions, ligand strain, ADMET properties, and model uncertainty.
 6. Select a final top-5 candidate list.
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Final Portfolio Statement

This project demonstrates that a REINVENT prior can be transfer-learned toward Factor Xa inhibitor-like chemical space using potent FXa seeds, and that a scaffold-aware ML model can quantify the shift in generated molecule quality. Compared with the original prior, the TL model increased the rate of high-scoring predicted FXa molecules and produced a novel, druglike candidate queue for docking follow-up.

This provides a clear bridge from ligand-based machine learning to generative molecular design and structure-based validation.